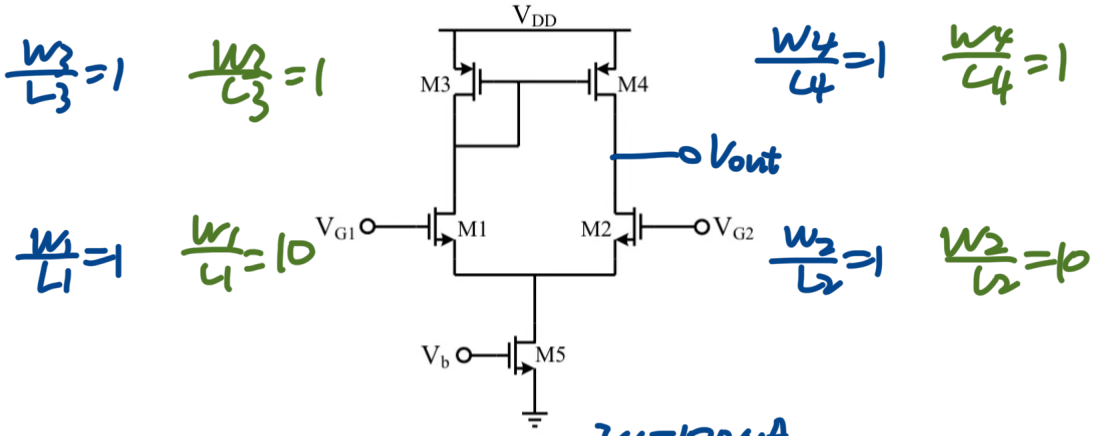
Homework #8

8.1 Calculate the differential transconductance g_{md} and the differential voltage gain A_v of an n-channel input differential amplifier shown in Figure 8.1 , with the parameters shown in table 8.1. Consider I_{ss}=100μA (the drain current of M5), and $W_1/L_1=W_2/L_2=W_3/L_3=W_4/L_4=1$. Assuming all the channel lengths are equal to 1μm, and V_{DD}=5V. If $W_1/L_1=W_2/L_2=10W_3/L_3=10W_4/L_4=10$, repeat the calculation

$V_{DD}=5V$



$\frac{W_3}{L_3}=1$ $\frac{W_4}{L_4}=1$ $\frac{W_1}{L_1}=1$ $\frac{W_2}{L_2}=1$ $\frac{W_1}{L_1}=10$ $\frac{W_2}{L_2}=10$

$L=1\mu m$

$K_n=134\mu A/V^2$

Figure 8.1 $I_{SS}=100\mu A$

解: $I_D=\frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$

$g_m=\frac{\partial I_D}{\partial V_{GS}}=\mu C_{ox}\frac{W}{L}(V_{GS}-V_{TH})=\sqrt{2\mu C_{ox}\frac{W}{L}I_D}$

$g_{md}=\sqrt{2\mu_n C_{ox}(\frac{W}{L})_1\frac{I_{SS}}{2}}=0.116mS$

$r_{o2}=\frac{1}{\lambda_n\frac{I_{SS}}{2}}=0.2M\Omega$, $r_{o4}=\frac{1}{\lambda_p\frac{I_{SS}}{2}}=0.1M\Omega$

故 $A_v=g_{md}(r_{o2}\|r_{o4})=7.33$

同理, $g_{md}=\sqrt{2\mu_n C_{ox}(\frac{W}{L})_1\frac{I_{SS}}{2}}=0.366mS$

$A_v=g_{md}(r_{o2}\|r_{o4})=24.4$

8.2 Calculate the maximum(V_{IC}(max)) and the minimum input common-mode voltages (V_{IC}(min)), and the input common mode voltage range (ICMR) of an n-channel input differential amplifier shown in Figure 8.1, with the parameters shown in table 8.1. Assume all MOSFETs are in saturation, all the $(W/L)_i$ are equal to 10μm/1μm, I_{SS}=10μA, and V_{DD}=5V.

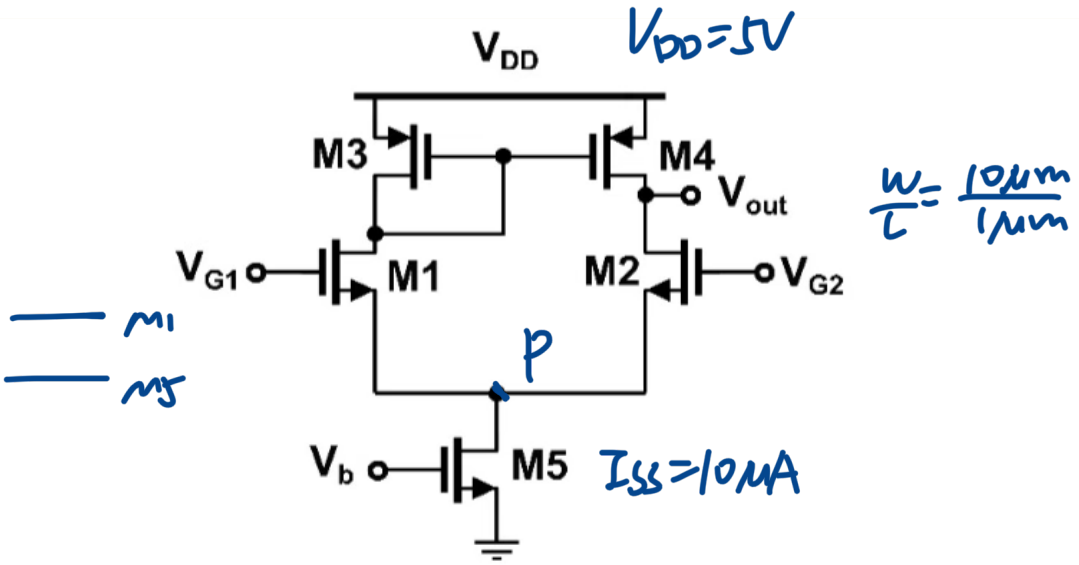


Figure 8.1

解: $V_{in,min} - V_{GS1} = V_{OV5} \Rightarrow V_{in,min} = V_{GS1} + V_{OV5}$

$V_{in,max} - V_{THn} = V_{DD} - V_{GS3} \Rightarrow V_{in,max} = V_{DD} - V_{GS3} + V_{THn}$

$I_1 = I_3 = \frac{1}{2} I_{SS} = 5\mu A$, $I_5 = I_{SS} = 10\mu A$

由 $I = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$

$V_{OV} = V_{GS} - V_{TH} = \sqrt{\frac{2I}{\mu C_{ox}(W/L)}}$

故 $V_{GS1} = \sqrt{\frac{2I_1}{\mu_n C_{ox}(W/L)_1}} + V_{THn} = 0.786V$

$V_{GS3} = \sqrt{\frac{2I_3}{\mu_p C_{ox}(W/L)_3}} + V_{THp} = 0.941V$

$V_{OV5} = \sqrt{\frac{2I_5}{\mu_n C_{ox}(W/L)_5}} = 0.122V$

故 $V_{in,min} = 0.908V$ $V_{in,max} = 4.759V$

$\Rightarrow ICMR = V_{in,max} - V_{in,min} = 3.851V$

8.3 Find the value of the unloaded differential-transconductance, g_{md}, and the unloaded differential-voltage gain, A_v, for the p-channel input differential amplifier of Figure 8.2 when I_{SS}=10μA and I_{SS}=1μA. What is the slew rate of the differential amplifier if a 100 pF capacitor is attached to the output? Assuming $W_1/L_1=W_2/L_2=W_3/L_3=W_4/L_4=1$, and all the channel lengths are equal to 1μm. Use the transistor parameters of Table 8.1.

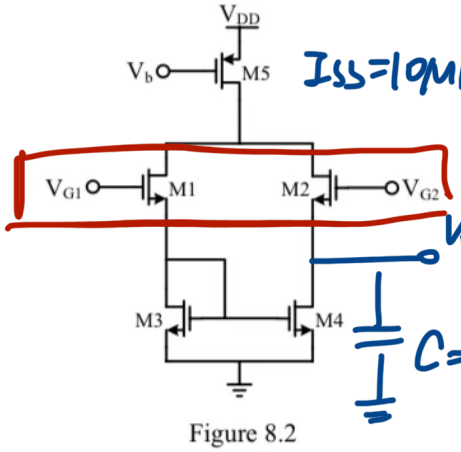


Figure 8.2

$I_{SS}=10\mu A$ $I_{SS}=1\mu A$

M1、M2会堵3，应该是P管

$\frac{W}{L}=1$ $L=1\mu m$

解 $I_D=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$

$g_m=\frac{\partial I_D}{\partial V_{GS}}=\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})=\sqrt{2\mu_n C_{ox}\frac{W}{L}I}$

故 $g_{md}=\sqrt{2\mu_p C_{ox}(\frac{W}{L})_1\frac{I_{SS}}{2}}=22.36\mu S$

$r_{o2}=\frac{1}{\lambda_p\frac{I_{SS}}{2}}=1M\Omega$ $r_{o4}=\frac{1}{\lambda_n\frac{I_{SS}}{2}}=2M\Omega$

故 $A_v=g_{md}(r_{o2}\|r_{o4})=14.91$

由 $I=C\frac{dV}{dt}$，$SR=\frac{dV_{out}}{dt}=\frac{I_{SS}}{C_L}=0.1V/\mu s$

同理，$g_{md}=\sqrt{2\mu_p C_{ox}(\frac{W}{L})_1\frac{I_{SS}}{2}}=7.07\mu S$

$r_{o2}'=\frac{1}{\lambda_p\frac{I_{SS}}{2}}=10M\Omega$，$r_{o4}'=\frac{1}{\lambda_n\frac{I_{SS}}{2}}=20M\Omega$

$A_v=g_{md}(r_{o2}'\|r_{o4}')=47.13$

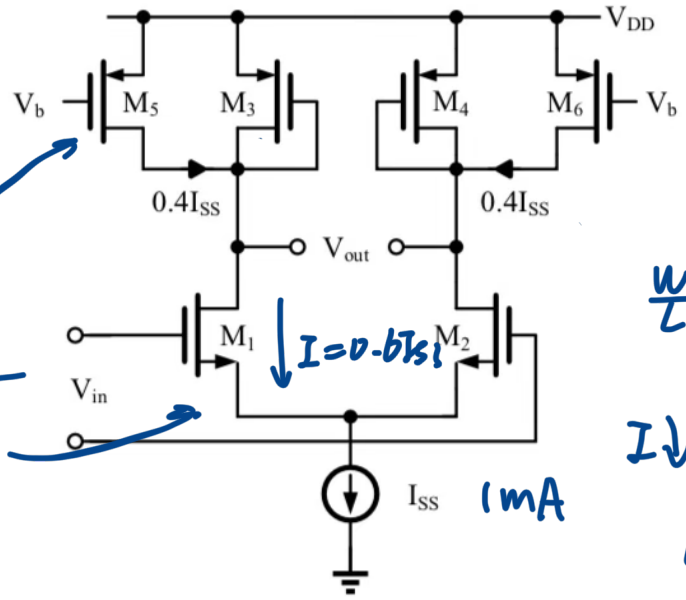
$SR=\frac{I_{SS}}{C_L}=0.01V/\mu s$

8.4 In the circuit of Fig. 8.3, assume that I_{SS}=1mA, V_{DD}=3V and W/L=50/0.5 for all the transistors. And $I_{D5}=I_{D6}=0.8(I_{SS}/2)$. Assuming $\lambda \neq 0$.

(a) Determine the voltage gain.

(b) Calculate V_b.

(c) If I_{SS} requires a minimum voltage of 0.4V, what is the maximum differential output swing?



M5, M3

M1, ISS

$\frac{W}{L}=\frac{50}{0.5}$

I↓ g_m↓ $R_D=\frac{1}{g_m}$↑

$A=g_mR_D$↑

Figure 8.3

解. (a) $I_3=0.1I_{SS}=100\mu A$, $I_1=\frac{1}{2}I_{SS}=500\mu A$

$$g_{m3}=\sqrt{2\mu_p C_{ox}(\frac{W}{L})_3 I_3}=1mS$$

$$g_{m1}=\sqrt{2\mu_n C_{ox}(\frac{W}{L})_1 I_1}=3.661mS$$

$$A_v=-g_{m1}\frac{1}{g_{m3}}=-3.661$$

(b) $I_5=0.4I_{SS}=400\mu A$

$$I=\frac{1}{2}\mu_p C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$$

$$\Rightarrow V_{GS5}=\sqrt{\frac{2I_5}{\mu_p C_{ox}(W/L)_5}}+V_{THp}=1.2V$$

故 $V_b=V_{DD}-V_{GS5}=1.8V$

(c) $V_{DD}-V_{out,max1}\simeq V_{DD}-V_b-V_{THP}\Rightarrow V_{out,max1}=2.6V$

$V_{out,max2}=V_{DD}-V_{THP}\Rightarrow V_{out,max2}=2.2V$

故 $V_{out,max}=2.2V$

$$V_{out,min1} = V_{DD} - |V_{GS2}|_{I_3 = 0.2I_{SS}}$$

$$= V_{DD} - \left(\sqrt{\frac{2I_3}{\mu_p C_{ox}(W/L)_3}} + |V_{THP}|\right)$$

$$= 3V - 1.083V = 1.917V$$

$$V_{out,min2} - V_{Imin} = (V_{GS1} - V_{TH})|_{I_1 = 0.6I_{SS}}$$

$$\Rightarrow V_{out,min2} = \sqrt{\frac{2I_1}{\mu_n C_{ox}(W/L)_1}} + V_{Imin} = 0.699V$$

故 $V_{out,min} = 1.917V$

故 $(\text{Differential Output Swing})_{max}$

$$= 2\times(V_{out,max} - V_{out,min}) = 0.566V$$

8.5 The circuit shown in Figure 8.4 called a folded-current mirror differential amplifier and is useful for low values of power supply. Assume that all W/L values of each transistor is 100. Using the parameters shown in table 8.1,

a) Find the maximum input common mode voltage, V_{IC}(max) and the minimum input common mode voltage, V_{IC}(min). Keep all transistors in saturation for this problem.

b) What is the input common mode voltage range, ICMR?

c) Find the small signal voltage gain, v_{out}/v_{in}, if $v_{in} = v_1 - v_2$.

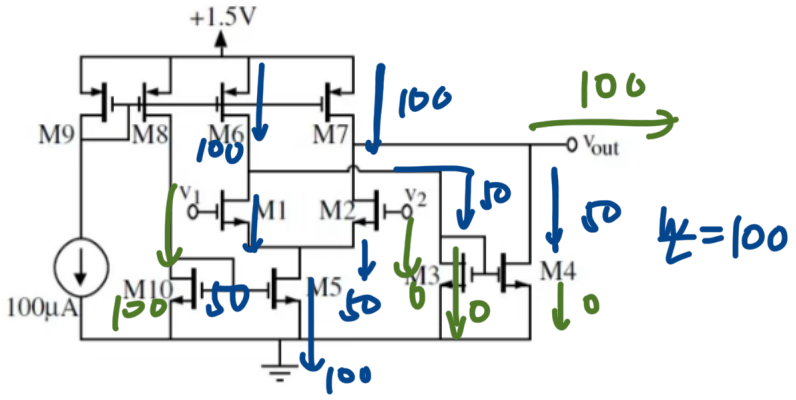


Figure 8.4

解. a) $I_1 = I_2 = I_3 = I_4 = I_5 = 50\mu A$, $I_6 = I_7 = 100\mu A$

$V_{IC(max)} - V_{THn} = V_{GS3}$

$\Rightarrow V_{IC(max)} = V_{GS3} + V_{THn}$

$= \sqrt{\frac{2I_3}{\mu_n C_{ox}(\frac{W}{L})_3}} + 2V_{THn} = 1.486V$

$V_{IC(min)} - V_{GS1} = V_{GS5} - V_{THn}$

$\Rightarrow V_{IC(min)} = V_{GS1} + V_{GS5} - V_{THn}$

$= \sqrt{\frac{2I_1}{\mu_n C_{ox}(\frac{W}{L})_1}} + \sqrt{\frac{2I_5}{\mu_n C_{ox}(\frac{W}{L})_5}} + V_{THn} = 0.908V$

(b) $V_{ICMR} = V_{IC(max)} - V_{IC(min)} = 0.578V$

(c) $r_{O2} = r_{O4} = \frac{1}{\lambda_n \frac{I_{SS}}{2}} = 0.2M\Omega$ $r_{O7} = \frac{1}{\lambda_p I_{SS}} = 0.05M\Omega$

$g_{m1} = \sqrt{2\mu_n C_{ox}(\frac{W}{L})_1 \frac{I_{SS}}{2}} = 1157.58\mu S$

故 $A_V = g_{m1}(r_{O2} \| r_{O4} \| r_{O7}) = 38.59$

8.6 In the circuit of Fig 8.5, assume that I_{SS} = 0.5mA, V_{DD} = 3V, $(W/L)_{1,2}$ = 50/0.5 and $(W/L)_{3,4}$ = 10/0.5. I_{SS} current is provided by NMOS, and its W/L = 50/0.5. Assuming $\lambda \neq 0$.

a) Calculate the range of input common mode voltage.

b) If $V_{in,CM}$ = 1.5V, draw a sketch of the small signal differential voltage gain of the circuit when V_{DD} changes from 0 to 3V.

c) If the mismatch threshold voltage of M_1 and M_2 is 1mV, calculate CMRR.

d) If the W_3 = 10μm and W_4 = 11μm, calculate CMRR.

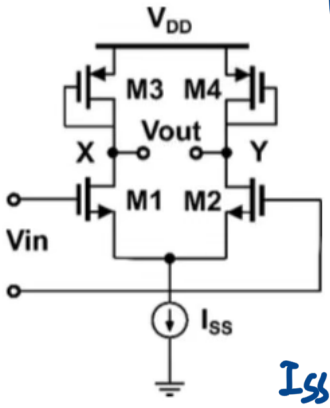


Figure 8.5

$V_{DD}=3V$

$(\frac{W}{L})_{3,4}=\frac{10}{0.5}$

$(\frac{W}{L})_{1,2}=\frac{50}{0.5}$

$I_{SS}=0.5mA$ $(\frac{W}{L})_2=\frac{50}{0.5}$

解. (a) $I=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$

$\Rightarrow V_{OV1}=\sqrt{\frac{2I_{SS}}{\mu_n C_{ox}(\frac{W}{L})_1}}=0.273V$

$V_{GS1}=\sqrt{\frac{2I_{SS}/2}{\mu_n C_{ox}(\frac{W}{L})_1}}+V_{THn}=0.893V$

$V_{in,min}=V_{GS1}+V_{OV1}=1.166V$

$V_{GS3}=\sqrt{\frac{2I_{SS}/2}{\mu_p C_{ox}(\frac{W}{L})_3}}+V_{THp}=1.507V$

$V_{in,max}-V_{TH}=V_{DD}-V_{GS3}$

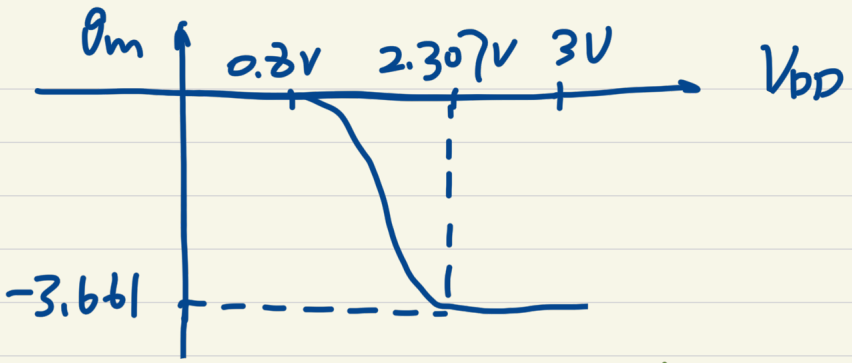
$\Rightarrow V_{in,max}=V_{DD}-V_{GS3}+V_{THn}=2.193V$

$1.166V \le ICMR \le 2.193V$, $\Delta V_{in}=1.027V$

(b) $V_{in,cm}=1.5V$, $V_{DD,min}=V_{in,cm}-V_{THN}+V_{GS3}=2.307V$

$g_{m3}=\sqrt{2\mu_p C_{ox}(\frac{W}{L})_3 I_3}=707.11\mu S$ $g_{m1}=\sqrt{2\mu_n C_{ox}(\frac{W}{L})_1 I_1}=2588.4\mu S$

故 $A_V=-g_{m1}\frac{1}{g_{m3}}=-3.661$



(c) 如本次作业最后附录中推导情况③

$$|A_{cm-dm}| = \frac{-g_{m1}\frac{1}{g_{m3}} + g_{m2}\frac{1}{g_{m4}}}{1+(g_{m1}+g_{m2})R_{SS}}$$

$$\frac{g_{m1}}{g_{m3}} = \frac{\sqrt{2\mu C_{ox}(\frac{W}{L})_1 I_1}}{\sqrt{2\mu_n C_{ox}(\frac{W}{L})_3 I_1}} = \sqrt{\frac{(\frac{W}{L})_1}{(\frac{W}{L})_3}} = \sqrt{\frac{(\frac{W}{L})_2}{(\frac{W}{L})_4}} = \frac{g_{m2}}{g_{m4}}$$

故 $A_{cm-dm}=0$，$CMRR = \left|\frac{A_{dm-dm}}{A_{cm-dm}}\right| = \infty$

(d) 如本次作业最后附录中推导情况①

$$|A_{cm-dm}| = \frac{g_m(\frac{1}{g_{m3}} - \frac{1}{g_{m4}})}{1+2g_m R_{SS}}$$

$$g_m = \sqrt{2\mu_n C_{ox}(\frac{W}{L})_{1,2}\frac{I_{SS}}{2}} = 2588.44\mu S$$

$$g_{m3} = \sqrt{2\mu_p C_{ox}(\frac{W}{L})_3\frac{I_{SS}}{2}} = 707.11\mu S$$

$$g_{m4} = \sqrt{2\mu_p C_{ox}(\frac{W}{L})_4\frac{I_{SS}}{2}} = 741.62\mu S$$

$$R_{SS} = \frac{1}{\lambda_n I_{SS}} = 0.02M\Omega$$

故 $|A_{cm-dm}| = 1.629\times10^{-3}$

由(b) $|A_{dm-dm}| = 3.661$

故 $CMRR = \left|\frac{A_{dm-dm}}{A_{cm-dm}}\right| = 2247.4$

8.7 A two-pole feedback system is designed such that $|\beta H(\omega_{p2})| = 1$ and $|\omega_{p1}| << |\omega_{p2}|$ (Figure 8.6). How much is the phase margin?

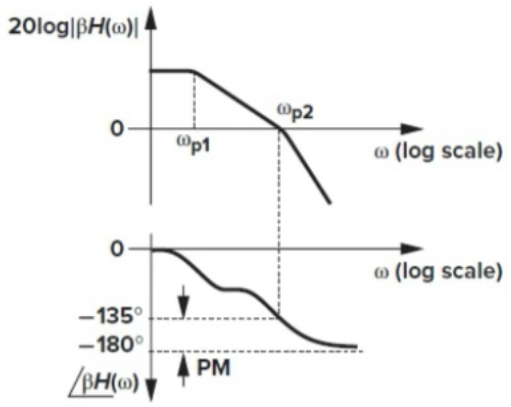


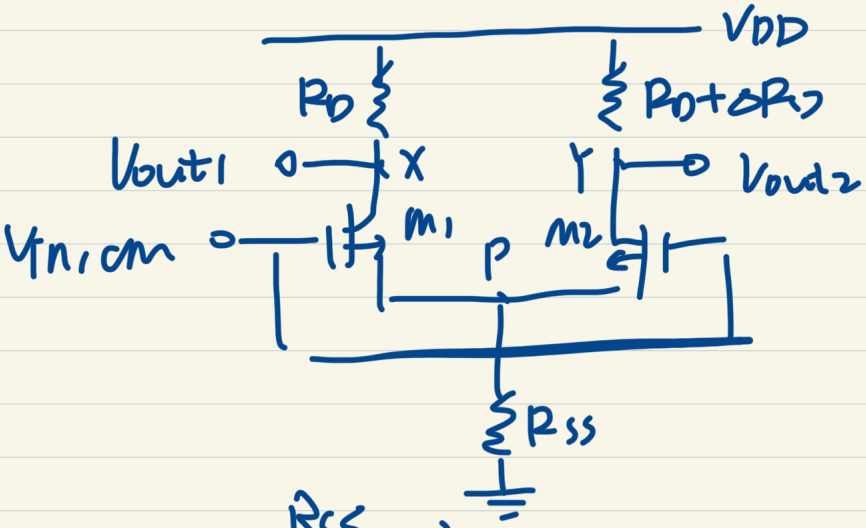
Figure 8.6

解. 在 $\omega = \omega_{p2}$ 处 $\angle \beta H = -135^\circ$

故 Phase Margin $= 45^\circ$

附录：A_{cm-dm}推导

①



$$\Delta V_P = \frac{R_{SS}}{R_{SS}+\frac{1}{2g_m}}\Delta V_{in,cm}$$

$$\Delta V_{GS} = \Delta V_{in,cm} - \Delta V_P = 1 - \frac{R_{SS}}{R_{SS}+\frac{1}{2g_m}}$$

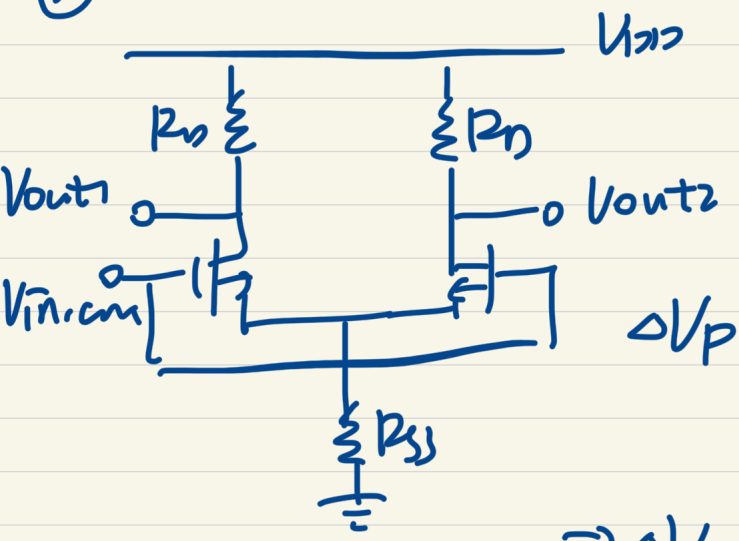
$$= \frac{\frac{1}{2g_m}}{R_{SS}+\frac{1}{2g_m}} = \frac{1}{1+2g_mR_{SS}}\Delta V_{in,cm}$$

$$\Delta V_X = -\Delta V_{in,cm}\frac{1}{1+2g_mR_{SS}}g_mR_D$$

$$\Delta V_Y = -\Delta V_{in,cm}\frac{1}{1+2g_mR_{SS}}g_m(R_D+\Delta R_D)$$

$$|A_{dm-cm}| = \left|\frac{\Delta V_X - \Delta V_Y}{\Delta V_{in,cm}}\right| = \frac{g_m}{1+2g_mR_{SS}}\Delta R_D$$

②



$$\Delta I_{D1} = g_{m1}(\Delta V_{in,cm} - \Delta V_p)$$

$$\Delta I_{D2} = g_{m2}(\Delta V_{in,cm} - \Delta V_p)$$

$$\Delta V_p = (\Delta I_{D1} + \Delta I_{D2}) R_{SS}$$

$$= (g_{m1} + g_{m2}) R_{SS} (\Delta V_{in,cm} - \Delta V_p)$$

$$\Rightarrow \Delta V_p = \frac{(g_{m1} + g_{m2}) R_{SS}}{1 + (g_{m1} + g_{m2}) R_{SS}} \Delta V_{in,cm}$$

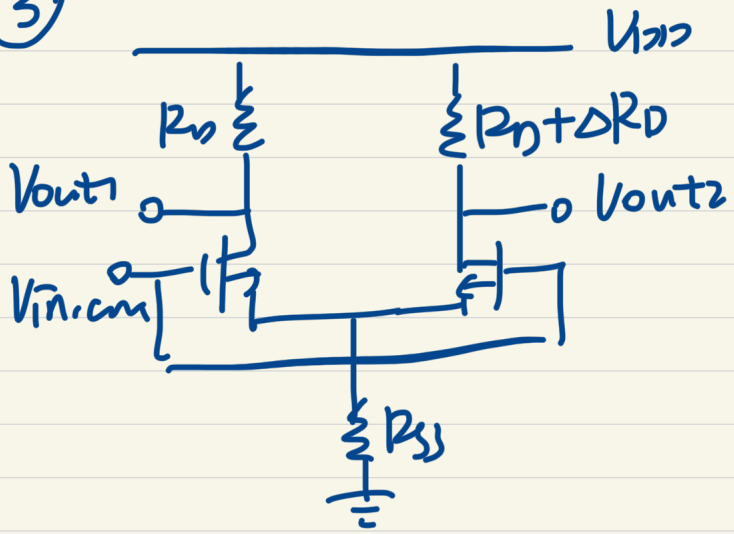
$$\Delta V_{GS} = \Delta V_{in,cm} - \Delta V_p = \frac{1}{1 + (g_{m1} + g_{m2}) R_{SS}} \Delta V_{in,cm}$$

$$\Delta V_x = -g_{m1} R_D \Delta V_{GS}, \quad \Delta V_y = -g_{m2} R_D \Delta V_{GS}$$

$$\text{故 } \Delta V_x - \Delta V_y = \frac{-(g_{m1} - g_{m2}) R_D}{1 + (g_{m1} + g_{m2}) R_{SS}} \Delta V_{in,cm}$$

$$|A_{cm-dm}| = \frac{(g_{m1} - g_{m2}) R_D}{1 + (g_{m1} + g_{m2}) R_{SS}}$$

③



$$\Delta V_x = -g_{m1} R_D \Delta V_{GS}, \quad \Delta V_y = -g_{m2}(R_D + \Delta R_D)\Delta V_{GS}$$

$$\therefore \Delta V_x - \Delta V_y = \frac{-g_{m1}R_D + g_{m2}(R_D + \Delta R_D)}{1 + (g_{m1} + g_{m2})R_{SS}} \Delta V_{in,cm}$$

$$|A_{cm-dm}| = \frac{-g_{m1}R_D + g_{m2}(R_D + \Delta R_D)}{1 + (g_{m1} + g_{m2})R_{SS}}$$